Session 5: Advanced Database Search & Profile Methods

KB8029 — Bioinformatics & Deep Learning

Arne Elofsson

SciLifeLab

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Log in to the lecture quiz

Log in to the quiz

Scan the code, or go to quiz.bioinfo.se/live

Log in with your university account

Answering counts as being at the lecture: up to 3 bonus points on the exam

Right or wrong is never graded, only whether you answered. Results are shown without names.

quiz.bioinfo.se/live

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Warm-up: what do you know already?

3 quick questions

On the topics of today, one at a time, about 30 seconds each

Answer on your own, and guess if you do not know

It shows me where to start; right or wrong is never graded

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BLAST reports a hit with E-value 0.001. What does that mean?

A The hit is homologous with 99.9% probability

B About 0.001 hits this good are expected by chance

C The sequences differ at 0.1% of their positions

D The alignment has 0.001 gaps per residue

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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In one column of an alignment, 4 of 5 sequences have A. Without pseudocounts, what is the frequency of A?

A 4

B 0.25

C 0.2

D 0.8

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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In a Markov chain, the next state depends on …

A only the current state

B all the earlier states

C nothing: it is random

D the final state

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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What the room knows already

Where we start

Which topics most of you know, and which we take slowly today.

Show the results ▸ quiz.bioinfo.se/live/report

Your questions before today

What you asked

The questions you sent in on Canvas before the lecture

The most common ones now; the others in the block where they belong

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In up to three words: what do you want to learn today?

BLAST	PSI-BLAST, PSSMs	Profile HMMs
seeds, extension, E-values	counts, pseudocounts, log-odds	match, insert, delete; Pfam domains

Answer now

Scan the code, or go to
quiz.bioinfo.se/live



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What you want to learn today

What the room wants to learn

Show the results ▸ quiz.bioinfo.se/live/report

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From one sequence to a model of the family



BLAST scores every position of the query with the same matrix. A PSSM gives every position its own substitution scores. A profile HMM adds where insertions and deletions are common.

See course.bioinfo.se/session05.html.

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Three blocks: about 30 min talk and 15 min discussion



Block 1: BLAST, how it searches and how to read its E-values. Block 2: PSI-BLAST and the position-specific scoring matrix (PSSM). Block 3: profile hidden Markov models.

Times are approximate: a 10 min start (warm-up quiz and your questions), two 15 min breaks and a 5 min close. Each block ends with a quiz question: guess, run the notebook and work out why it behaves as it does, discuss, answer again.

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BLOCK 1 · ABOUT 30 MIN

BLAST: seed, extend, evaluate

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Optimal alignment does not scale to a database

- **DP is optimal, but slow per pair**
- Times a hundred million sequences
- **BLAST: speed instead of a guarantee**

$m \times n$

cells per pair
(Session 4)

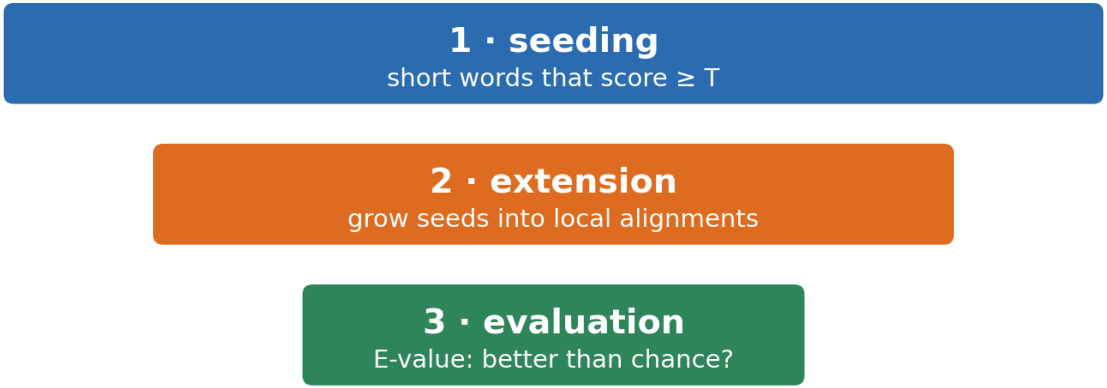
$> 10^8$

sequences in GenBank
and UniProt

See course.bioinfo.se/session05.html, 'The scaling problem, and BLAST's heuristic answer'.

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Three stages, each passing on less



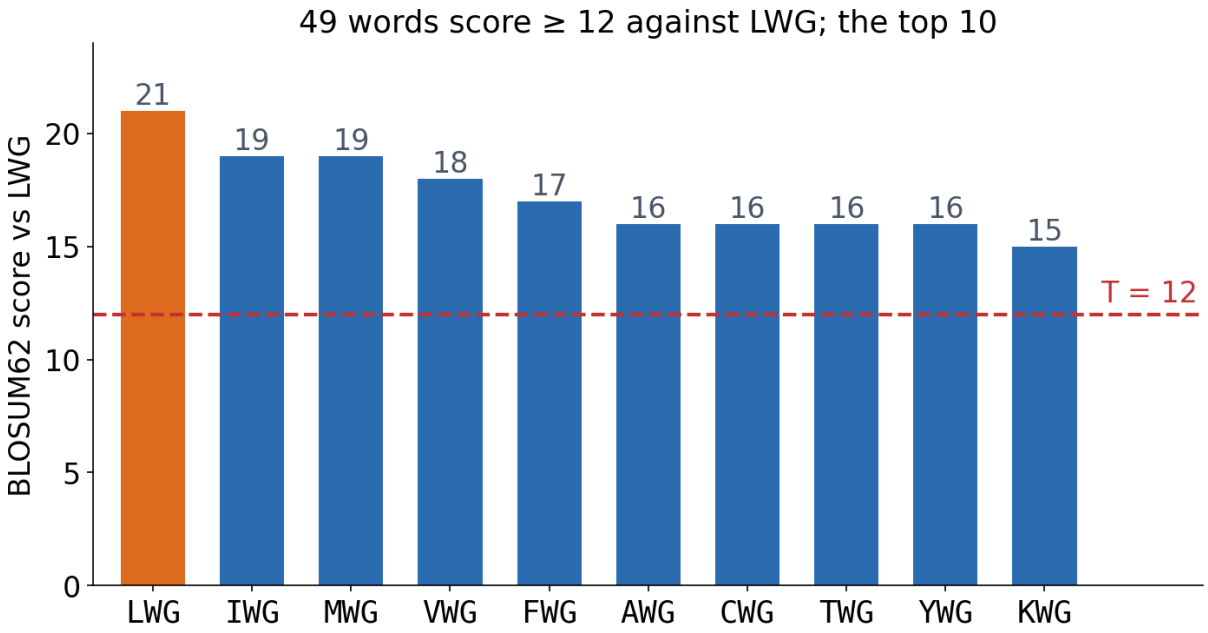
from every database sequence to a few alignments worth reporting

- **Seeding: find short matches**
- **Extension: grow them**
- **Evaluation: beat chance?**
- Real BLAST adds refinements

See course.bioinfo.se/session05.html, 'The scaling problem, and BLAST's heuristic answer'.

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Seeds: words that score at least T

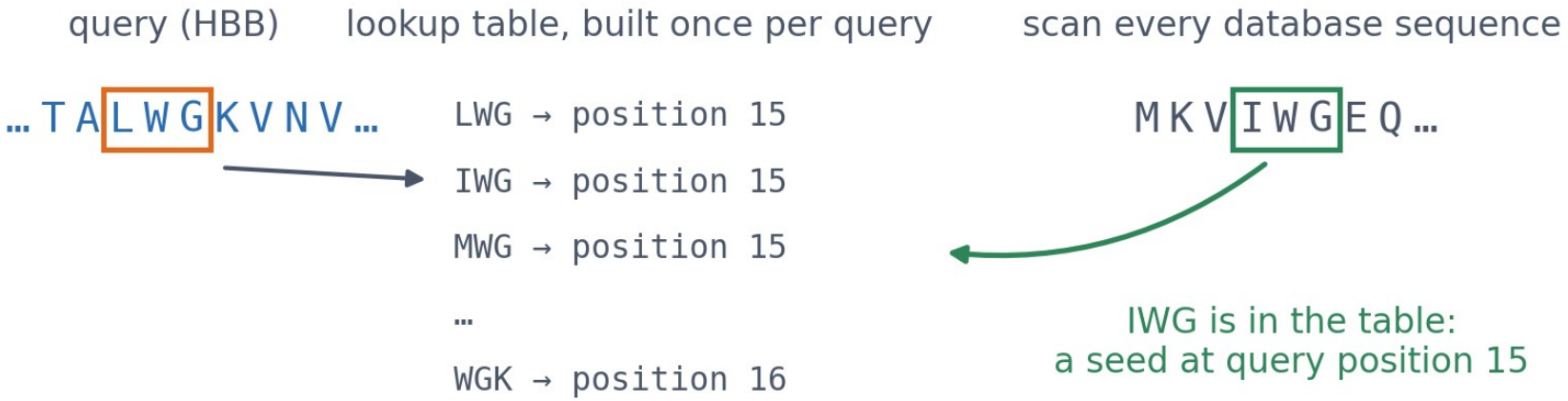


- **Protein words: 3 residues**
- Not only exact matches
- Scored with BLOSUM62
- **LWG: 49 words ≥ 12**

Neighbour words computed from BLOSUM62, as in the course notebook.

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The lookup table: built from the query, used while scanning



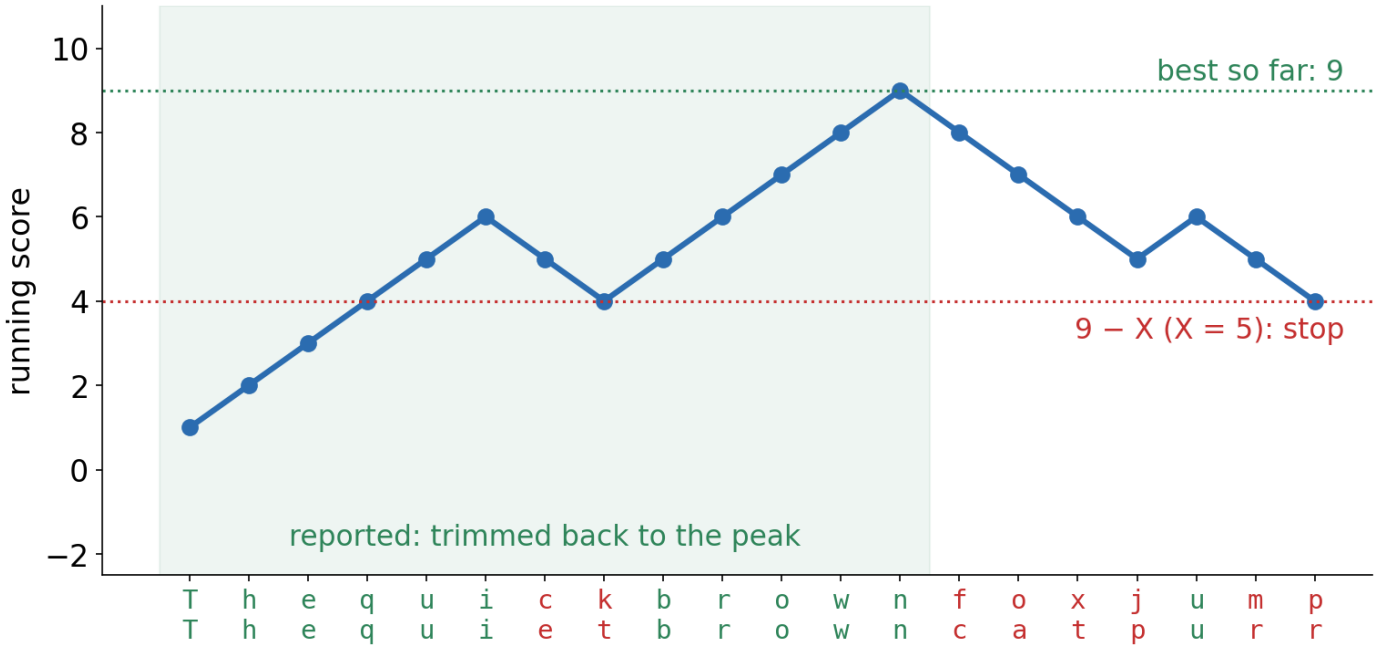
In HBB, LWG starts at position 15 (counting the initial Met)

Every query word and its high-scoring neighbours go into a table. BLAST then reads each database sequence once: one table lookup per word, so the scan is linear in the size of the database.

See course.bioinfo.se/session05.html, 'Stage 1: seeding'.

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Extension and the X-drop rule

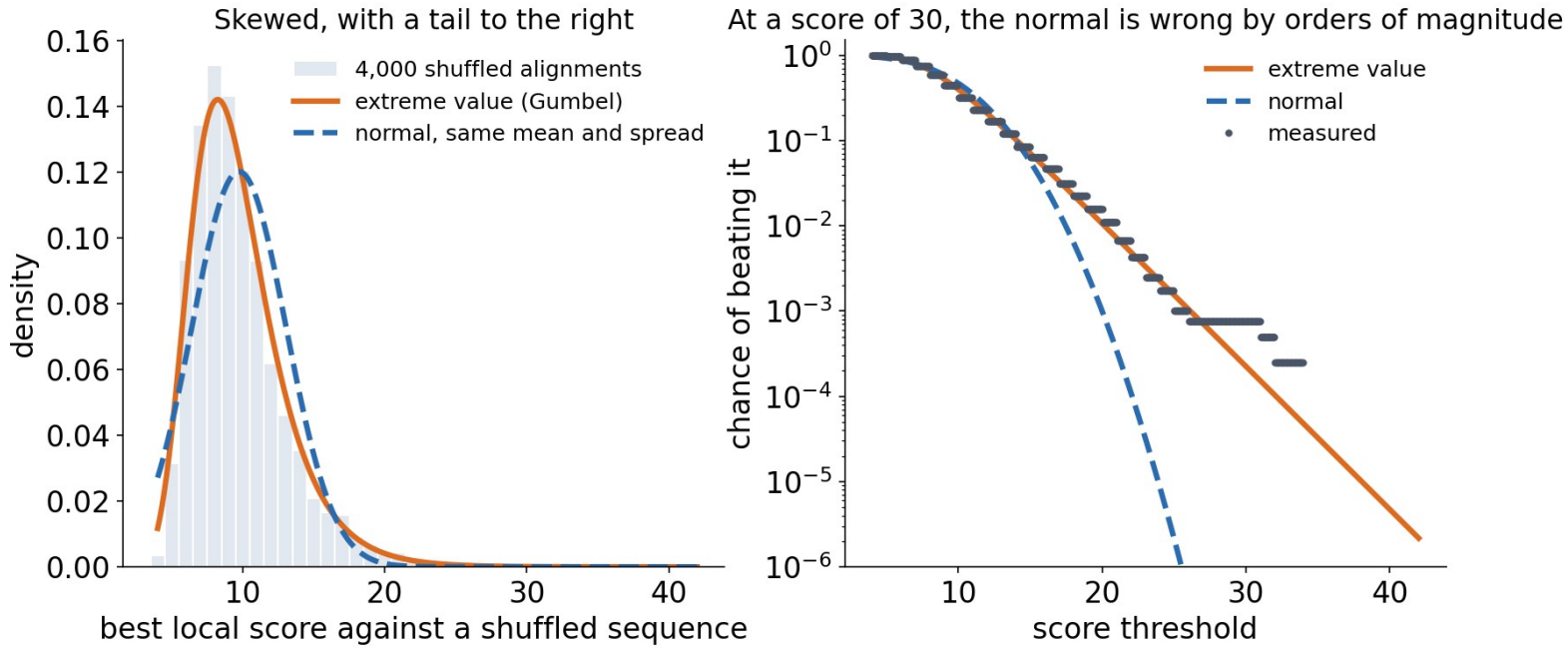


+1 for a match, −1 for a mismatch (spaces ignored). The running score peaks at 9 after 'brown'; once it is 5 below the peak, extension stops and the alignment is trimmed back to the peak: Thequickbrown / Thequietbrown.

Korf, Yandell & Bedell (2003); reproduced in the course notebook.

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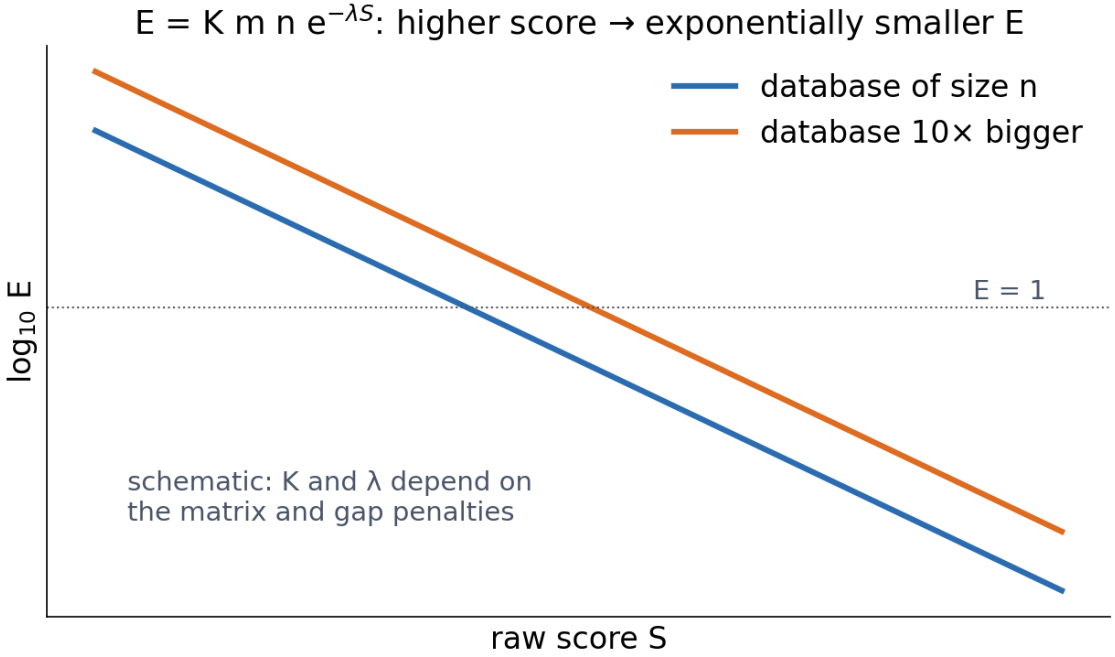
Why the E-value is not a p-value from a normal curve



4,000 shuffled alignments, scored. Left: the distribution is skewed, with a tail to the right. Right: the same data as the chance of beating a threshold. The extreme-value curve tracks the measurements for six orders of magnitude; the normal collapses.

Simulated and plotted by tools/builders/make_session05_slide_figures.py. Karlin & Altschul (1990), PNAS 87:2264.

The E-value: hits this good expected by chance



- **E = K m n e^(−λS)**
- Higher score: exponentially smaller E
- **Bigger database: bigger E**

Karlin & Altschul (1990); see course.bioinfo.se/session05.html, 'Stage 3'.

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Bit score or E-value?

Bit score

$$S' = (\lambda S - \ln K) / \ln 2$$

- Normalises away the scoring system
- Comparable across searches and matrices
- No universal cutoff

E-value

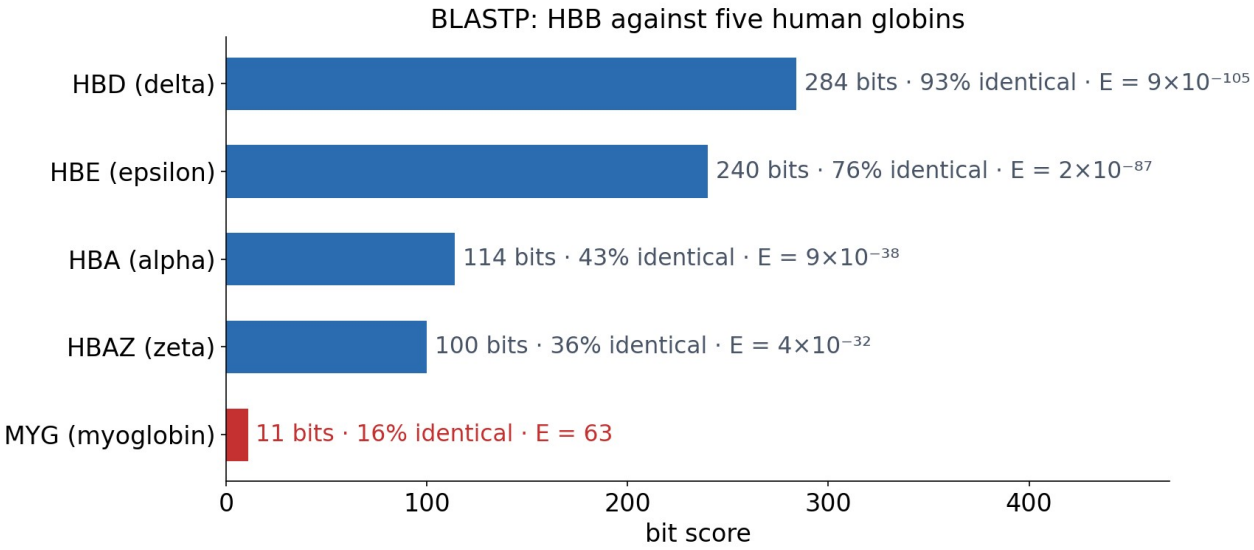
$$E = K\ m\ n\ e^{(-\lambda S)}$$

- Depends on the database size
- The one to read for significance
- E = 63: no evidence at all

See course.bioinfo.se/session05.html, 'Stage 3: is this hit significant?'.

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Reading real output: HBB against its relatives

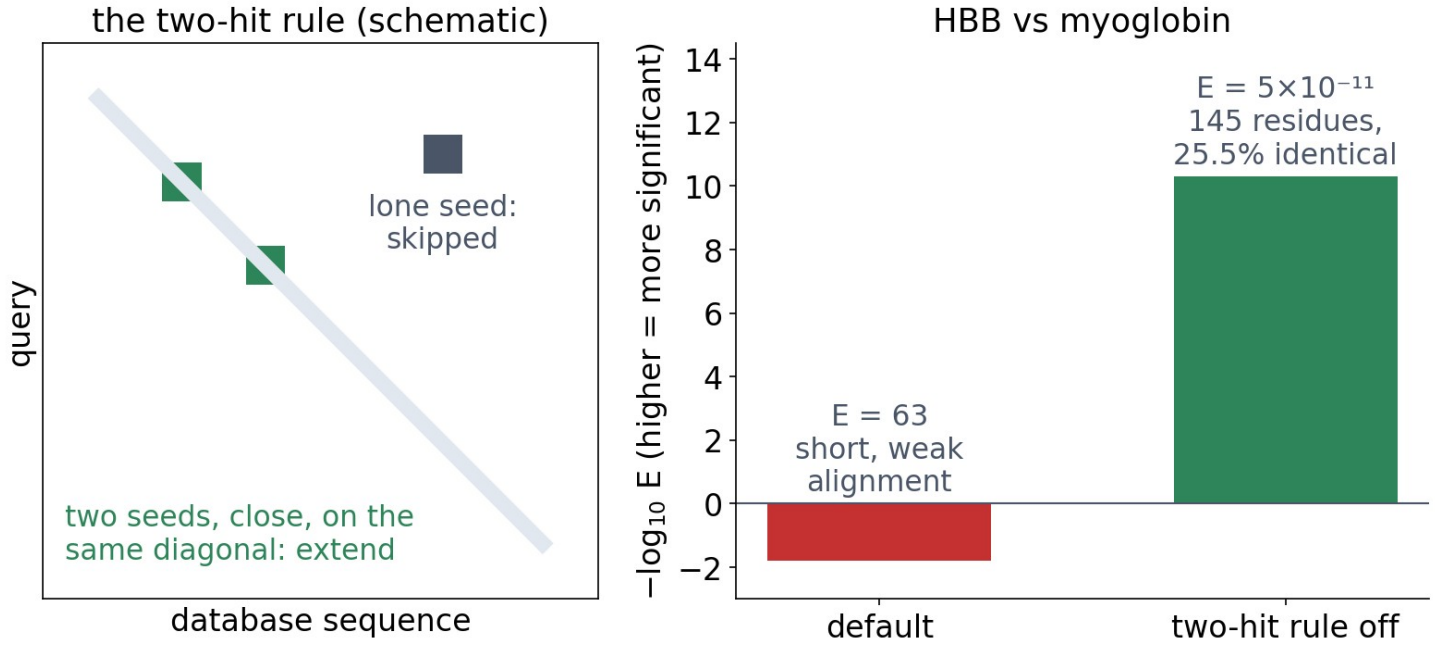


- **Four confident hits**
- **Myoglobin: E = 63**
- A real homolog, yet no evidence?

BLASTP (BLAST+) output in the course notebook; see course.bioinfo.se/session05.html, Table 5.2.

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The price of a heuristic: the two-hit rule



With the default settings no two seeds fell close enough together on the right diagonal, so the good alignment was never extended. Switched off (-window_size 0 in BLAST+), BLAST finds 145 residues at 25.5% identity, E = 5×10^{-11}.

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The same alignment, a bigger database

- **The same myoglobin alignment**
- Significant, but only just
- Truncated haemoglobins: no pairwise signal

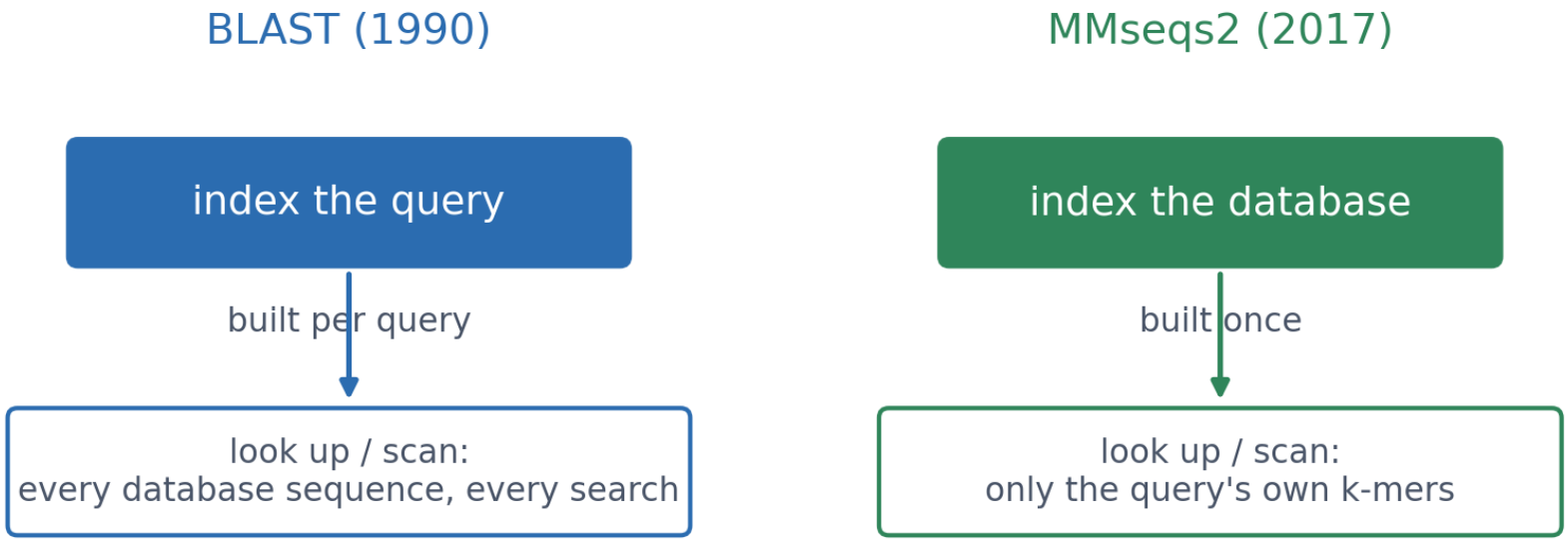
5×10^{-11}
against 5 globins

≈ 0.007
against UniProtKB ($\approx 9\times10^{10}$ residues)

See course.bioinfo.se/session05.html, 'Stage 3: is this hit significant?'.

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BLAST's ideas at today's scale: MMseqs2

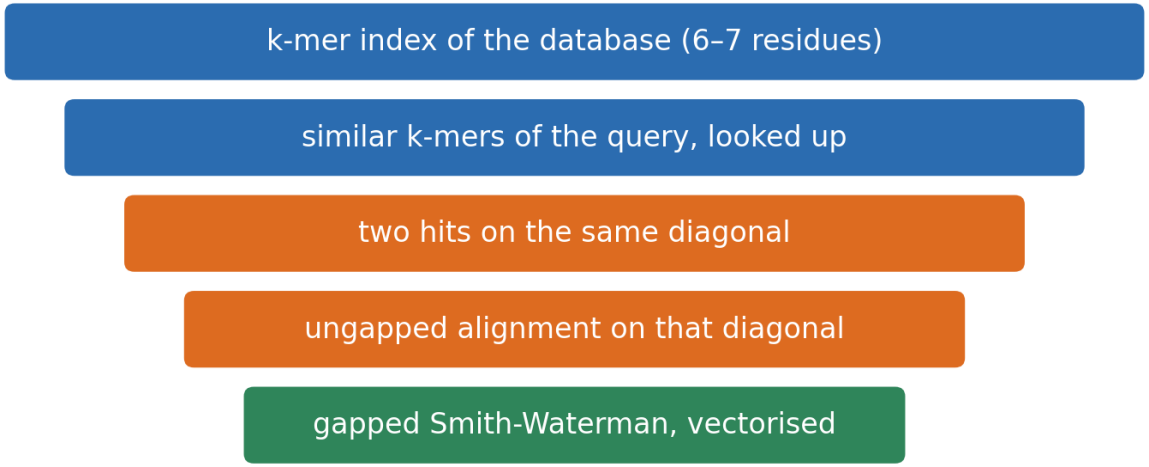


BLAST builds a table from the query and reads through every database sequence. MMseqs2 (Steinegger & Söding 2017) indexes the database once, so each query only looks up its own k-mers.

Steinegger & Söding (2017), Nature Biotechnology 35:1026-1028.

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A cascade of ever more expensive filters



- **Often hundreds of times faster**
- Similar sensitivity
- No machine learning
- Builds ColabFold's MSAs (Session 12)

See course.bioinfo.se/session05.html, 'BLAST's ideas at modern database scale: MMseqs2'.

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How many neighbour words does GKV have?

Neighbour words: every word scoring at least T = 12 against the query word with BLOSUM62. LWG has 49.

haemoglobin beta



BLOSUM62 diagonal (a residue against itself):

L–L +4 **W–W +11** G–G +6 K–K +5 V–V +4

LWG: 49 neighbour words at T = 12

GKV at the same T: ?

A About as many as LWG

B More: G, K and V have similar residues

C Far fewer: a lower self-score

D None: no word reaches 12

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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How many neighbour words does GKV have?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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How many neighbour words does GKV have?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session05-discussion-1.ipynb

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How many neighbour words does GKV have?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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How many neighbour words does GKV have?

Results, and who explains

Both rounds side by side.

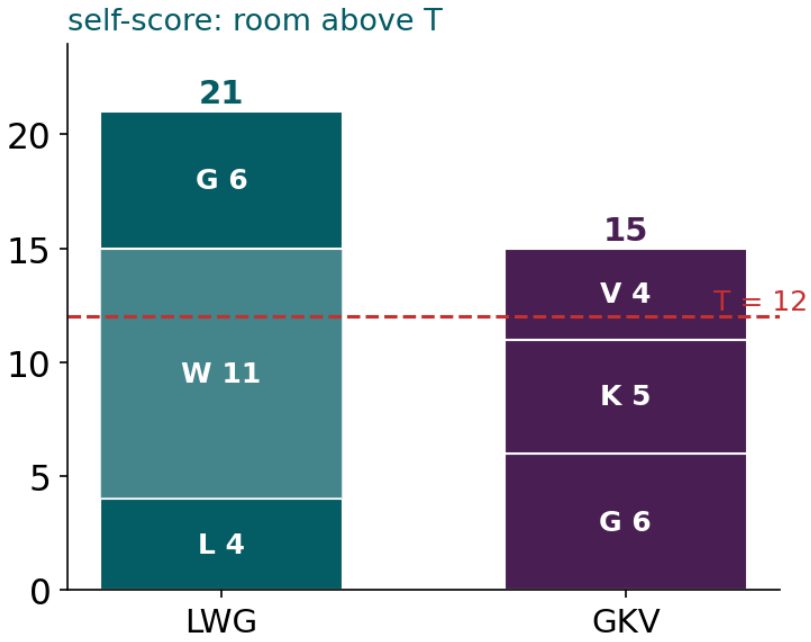
Draw two to explain: one of you says how you reasoned.

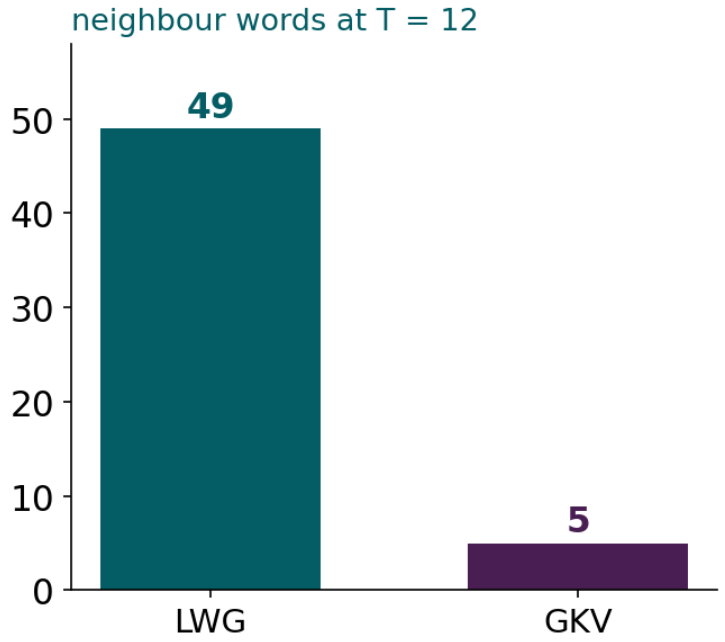
Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 1 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session05-discussion-1.ipynb

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The answer: GKV has only 5 neighbour words





LWG starts from a self-score of 21, mostly from W (+11), so many words with substitutions still reach T = 12. GKV starts from 15. The number of seeds depends on the query itself: rare, high-scoring residues give many; a lower T gives more, and a slower search.

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BLOCK 1 DONE

Break

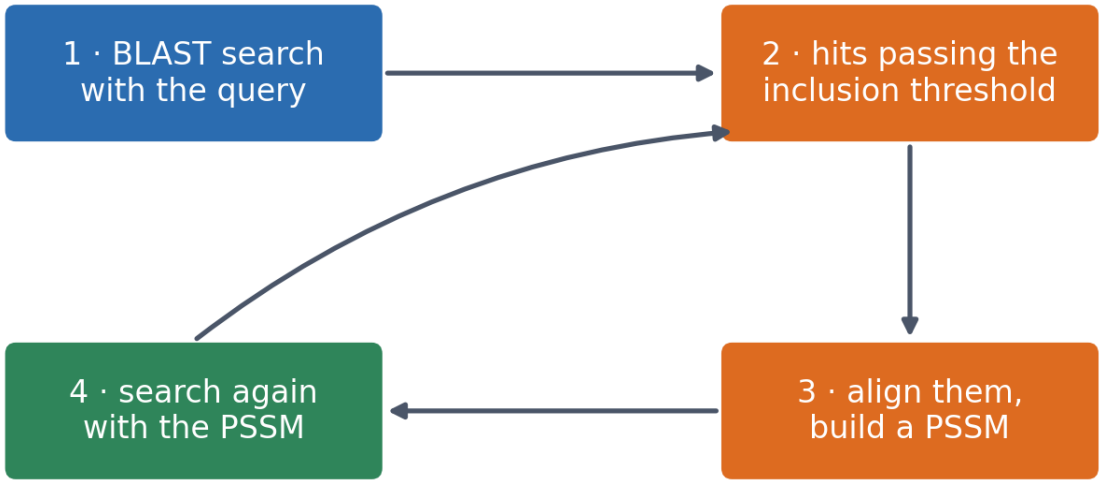
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BLOCK 2 • ABOUT 30 MIN

PSI-BLAST and the PSSM

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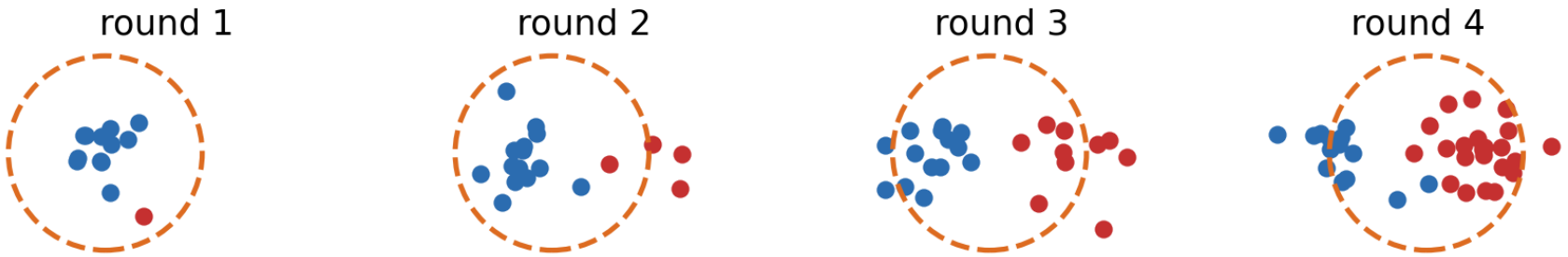
PSI-BLAST: BLAST, iterated with a profile



- **A scoring matrix for this query**
- Refined every round
- Finds remote homologs

See course.bioinfo.se/session05.html, 'From one sequence to many: PSI-BLAST and the PSSM'.

Iteration can go wrong: profile drift



blue: the family · red: an intruder and its relatives · orange: what the profile describes (schematic)

One unrelated sequence passes the inclusion threshold, shifts the next PSSM towards itself, and pulls in more like it. After a few rounds the 'family' can be dominated by proteins unrelated to the query.

See course.bioinfo.se/session05.html, 'Iteration is not automatically better: profile drift'.

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Step 1: count each residue at each position

	1	2	3	4	5
protein 1	A	G	L	S	P
protein 2	A	G	L	T	P
protein 3	R	G	I	S	P
protein 4	A	A	L	S	Q
protein 5	A	A	L	S	P

position 1: 4 × A, 1 × R · position 5: 4 × P, 1 × Q

- **A logo describes a column**
- **A PSSM scores a new sequence**
- Real tools weight sequences

The toy alignment of course.bioinfo.se/session05.html, 'Building a PSSM, step by step'.

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Step 2: add pseudocounts, a prior

- **A residue never seen gets frequency 0**
- Five sequences: absence may be luck
- Prior: every residue is possible
- Many sequences swamp the prior

−∞

log-odds of a zero frequency

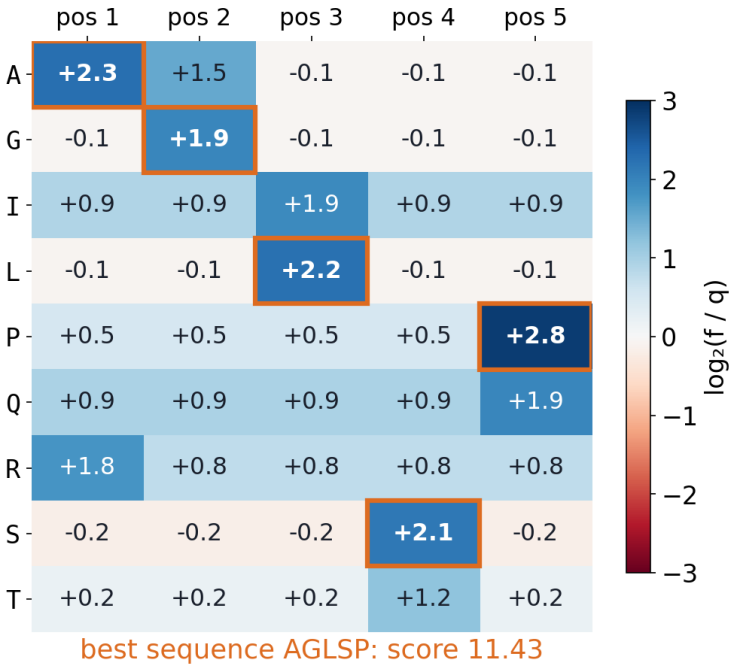
+1

added to every count (teaching model)

See course.bioinfo.se/session05.html, 'Building a PSSM, step by step'.

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Steps 3-4: compare with the background, take log-odds



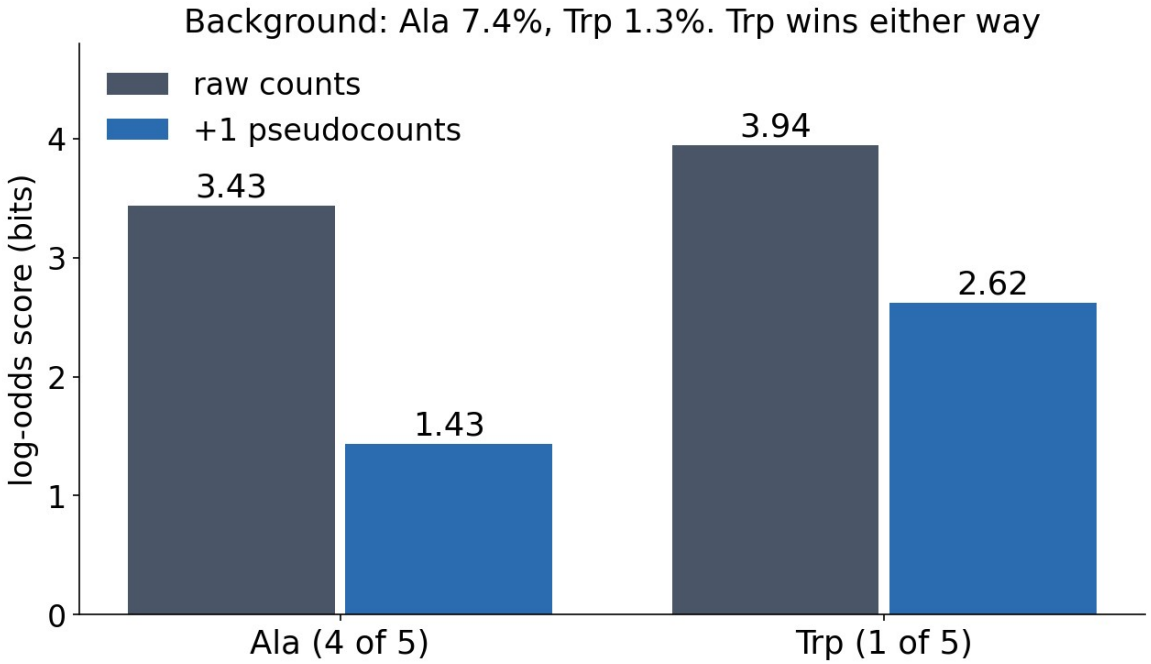
best sequence AGLSP: score 11.43

- **$S = \log_2(f / q)$**
- +1: twice as likely as by chance
- Best sequence: the consensus
- **AGLSP, 11.43**

PSSM as built in the course notebook (+1 pseudocounts over the residues in the alignment).

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Why pseudocounts matter, and what they cannot do

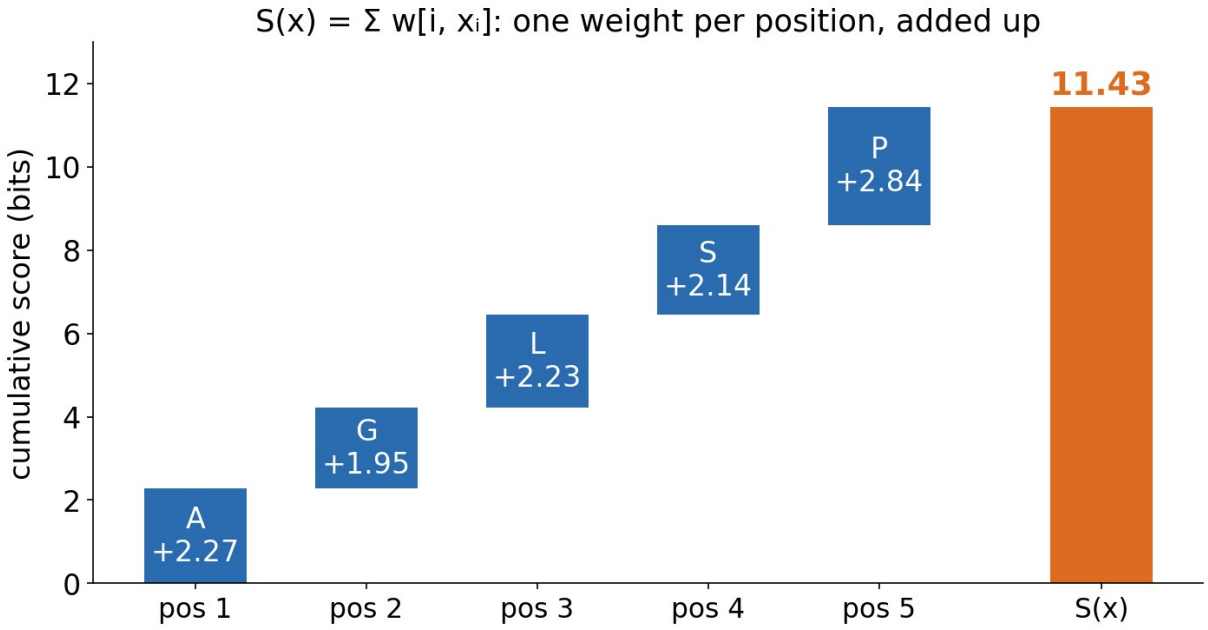


- **One rare Trp outscores four Ala**
- +1: both scores more cautious
- **The ranking stays**
- Only more sequences can tell

See course.bioinfo.se/session05.html, 'Why pseudocounts really matter'.

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A PSSM score is a sum: already a linear model



- **One weight per position**
- Look up, then add
- Session 7 learns these weights

PSSM of the course notebook; see course.bioinfo.se/session05.html, 'A PSSM is already a linear model'.

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What does an unseen residue do to the score?

PSSM from five aligned proteins, +1 pseudocount for every residue. GGLSP has a G at position 1, where none of the five has one.

position	1	2	3	4	5
protein 1	A	G	L	S	P
protein 2	A	G	L	T	P
protein 3	R	G	I	S	P
protein 4	A	A	L	S	Q
protein 5	A	A	L	S	P
new	G	G	L	S	P

pseudocount +1 for every residue: score vs AGLSP?

A Minus infinity: G was never seen there

B Lower than AGLSP, but finite

C The same: one position hardly matters

D Higher: G is common in the alignment

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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What does an unseen residue do to the score?

How did the room answer?

Halt the round when most have answered, then show the results.

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SciLifeLab

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quiz.bioinfo.se/live

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What does an unseen residue do to the score?

Results, and who explains

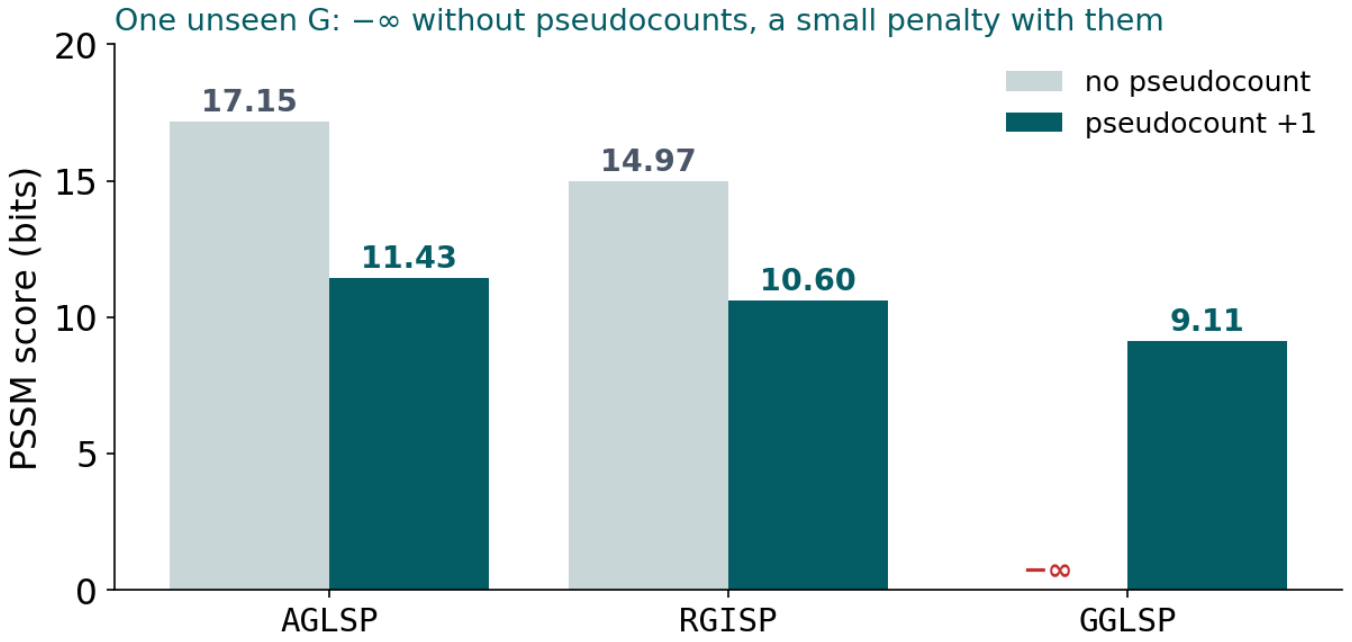
Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 2 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session05-discussion-2.ipynb

The answer: lower, but finite



Without pseudocounts, an unseen residue has frequency 0 and log-odds −∞: one position rules out the whole sequence. With +1 pseudocounts, GGLSP scores 9.11 against 11.43 for the consensus. Five sequences are too few to conclude that G can never occur at position 1.

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BLOCK 2 DONE

Break

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BLOCK 3 • ABOUT 30 MIN

Profile hidden Markov models

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A PSSM charges the same gap penalty everywhere

core: almost no gaps								loop: insertions common					core						
L	V	K	A	W	G	K	V	-	-	-	N	G	D	E	V	G	A	L	L
L	V	S	A	W	G	K	V	S	S	P	N	G	E	E	V	G	A	L	L
L	V	K	A	W	G	K	V	-	-	-	-	-	D	E	V	G	A	L	L
L	I	K	A	W	G	K	V	A	-	-	N	G	D	E	V	G	A	M	L

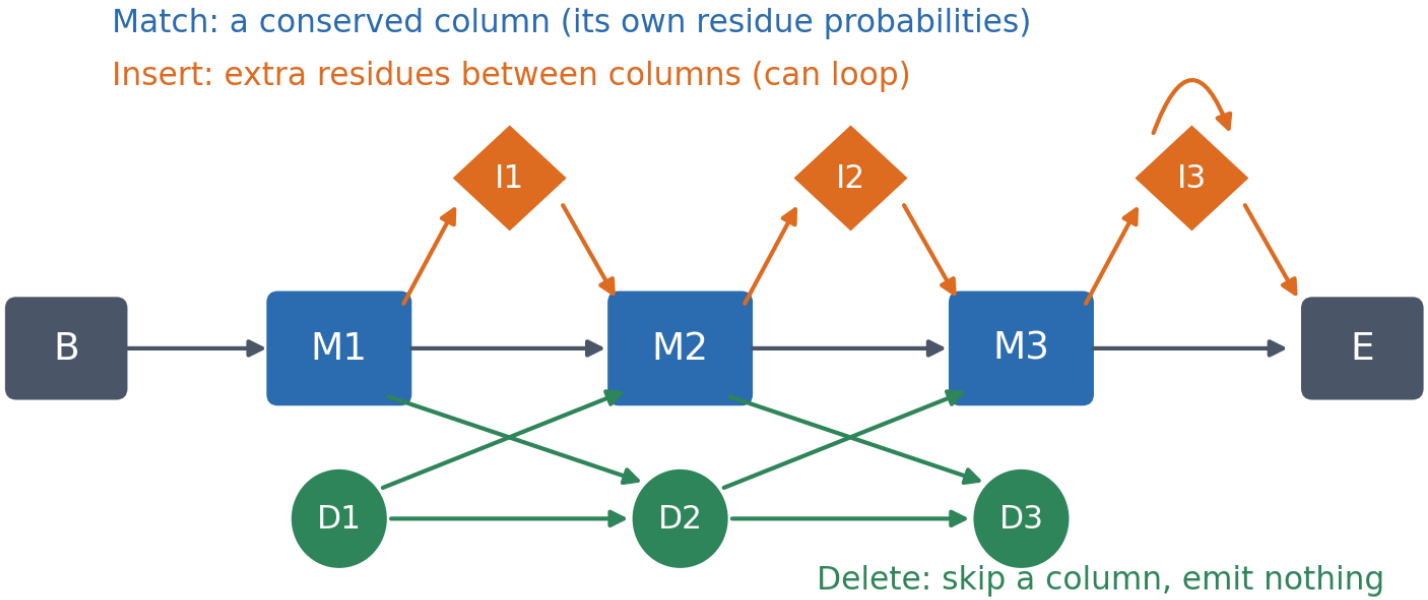
Schematic alignment. A PSSM charges the same gap penalty in both places

- **Loops tolerate insertions**
- **The core almost never**
- A model should know where

See course.bioinfo.se/session05.html, 'Profile Hidden Markov Models'.

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Match, insert and delete states, one set per position



Match states emit residues like one PSSM column; insert states emit extra residues and can loop; delete states skip a column and emit nothing. Every transition has its own learned probability.

See course.bioinfo.se/session05.html, 'Profile Hidden Markov Models'.

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Why "hidden"?



Same residues, different states. The model scores every path; the states stay hidden.

- **A residue does not say its state**
- Inferred from the best path
- Emissions + transitions together

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PSSM or profile HMM?

PSSM

PSI-BLAST

- Substitution scores per position
- Gaps: the same penalties everywhere
- Built during the search

Profile HMM

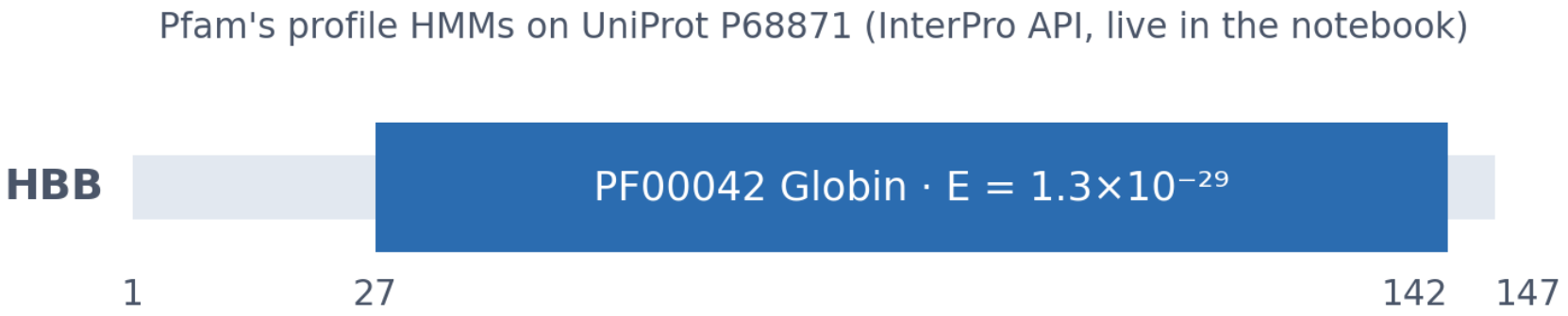
Pfam, HMMER

- Match, insert, delete per position
- Knows where gaps are common
- Trained on curated families

See course.bioinfo.se/session05.html, 'Profile Hidden Markov Models'.

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A real profile-HMM hit on HBB



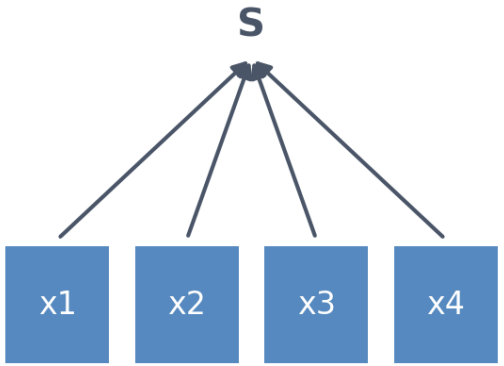
Pfam's profile HMMs, searched against UniProt P68871 (HBB_HUMAN): PF00042 'Globin' covers residues 27-142 of the 147, with E ≈ 1.3×10^{-29}. The same job the toy PSSM did on paper, at production scale.

Live InterPro API call in the course notebook.

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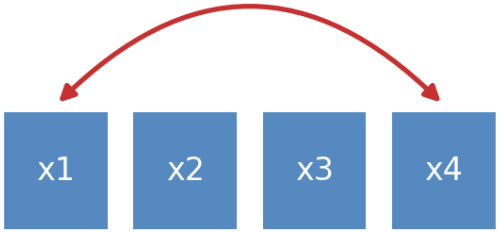
What neither a PSSM nor a profile HMM can represent

a profile: $S = w_1 + w_2 + w_3 + w_4$



epistasis: one position's effect depends on another

a pair, not a sum



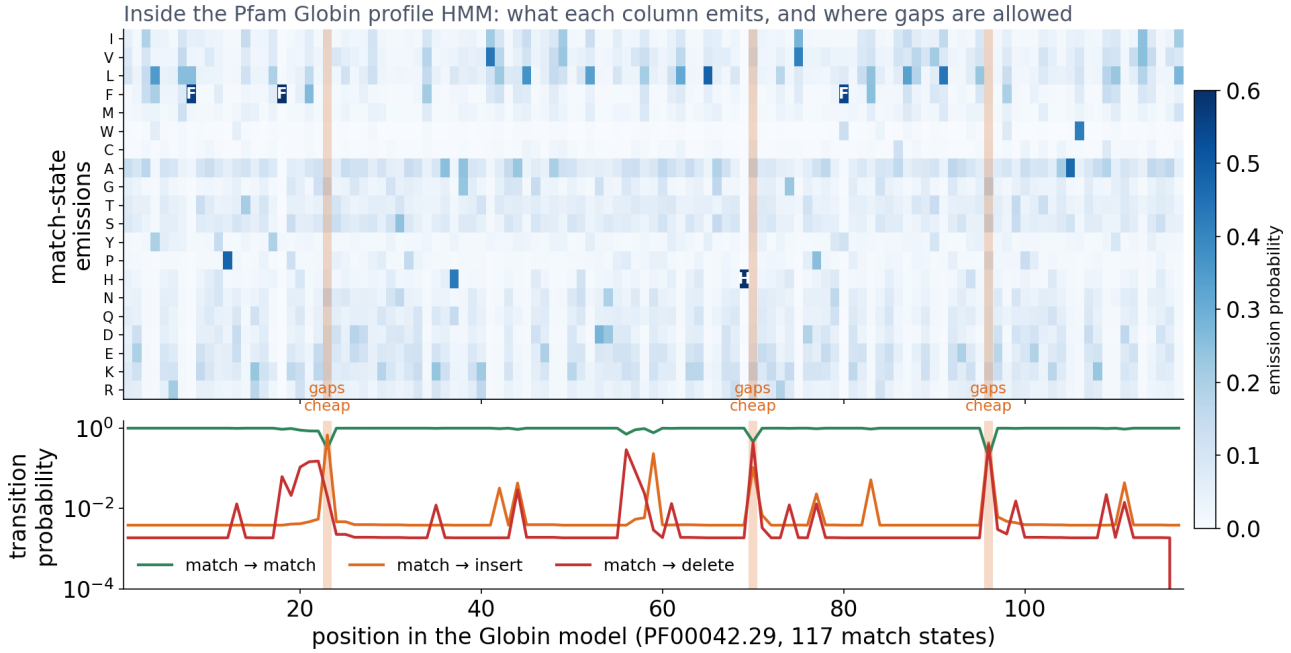
Session 9: all 160,000 variants at four positions of GB1 · Session 12: pairwise couplings

Both score a sequence position by position, so each position contributes independently. When the effect of one mutation depends on the residue at another position (epistasis), no profile can capture it.

See course.bioinfo.se/session05.html, 'What neither a PSSM nor a profile HMM can represent'.

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What a real profile HMM looks like inside: Pfam Globin



Top: for each of the 117 match states, the probability of each amino acid; a few columns are almost fixed (one His, three Phe). Bottom: transition probabilities; in most columns a gap is very unlikely, but at a few loop positions inserting or deleting costs almost nothing (orange).

Pfam PF00042 (Globin), fetched from the InterPro API (HMMER3 format, built from 73 seed sequences).

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From a profile you can read to a vector you cannot

a profile HMM you can read

one column per position

emission and gap probabilities

a biologist can check it

an embedding you cannot

hundreds of numbers per protein

learned by a language model

compared by vector distance

- Databases grow faster than CPUs
- MMseqs2 on GPUs
- Embeddings replace profiles in search
- **What do we lose when a profile becomes a number we can't read?**

search at scale: MMseqs2 on GPUs (2025), one search 6× faster than on 128 CPU cores

what do we lose when a profile becomes a number we can't read?

Kallenborn et al. (2025), Nature Methods, doi:10.1038/s41592-025-02819-8 (MMseqs2-GPU); embedding search: Session 4's outlook and Session 11.

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More family information, more distant relatives



more family information → more distant relatives found

BLAST: fast, one matrix for every position, E-values tied to the database. PSI-BLAST: a PSSM refined by iteration, with the risk of drift. Profile HMMs: position-specific gaps too, the standard behind Pfam.

See course.bioinfo.se/session05.html, 'What's next'.

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Which BRCA1 domain match is the most significant?

Three real Pfam domains: a RING finger (41 residues), a serine-rich domain (164) and a BRCT domain (79). Which has the smallest E-value?

BRCA1, 1863 residues: three real Pfam domains



Which match has the smallest E-value?

A The RING finger

B The BRCT domain

C The serine-rich domain

D About the same: all are real

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

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Which BRCA1 domain match is the most significant?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Which BRCA1 domain match is the most significant?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session05-discussion-3.ipynb

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Which BRCA1 domain match is the most significant?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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Which BRCA1 domain match is the most significant?

Results, and who explains

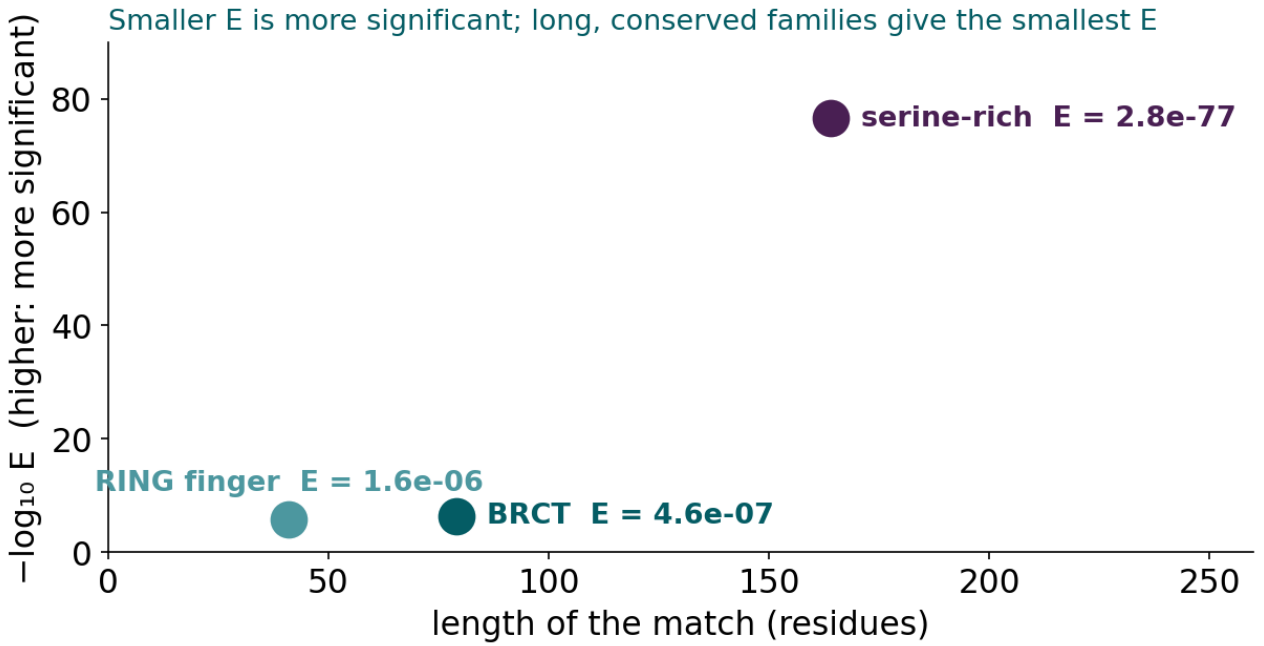
Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

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The answer: the serine-rich domain, E = 2.8×10^{-77}



Smaller E is more significant (for a bit score, bigger is better). A long domain adds up score over many positions, and a family of close relatives gives a specific profile. The short RING finger, from a diverse family, gets only 1.6×10^{-6}: weaker evidence from sequence alone, not a sign that the domain is absent.

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In up to three words: what was the hardest part of today?

BLAST	PSI-BLAST, PSSMs	Profile HMMs
seeds, extension, E-values	counts, pseudocounts, log-odds	match, insert, delete; Pfam domains

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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The hardest part of today

What the room found hardest

Show the results ▸ quiz.bioinfo.se/live/report

Session 6 next

From sequence to structure: the four levels, hands-on in PyMOL
Then Session 7: a PSSM score $S(x) = \Sigma\, w[i, x_i]$ is already a linear model
And its weights can be learned, not only derived from alignment statistics